

Individual Project Reflection

What?

This project allowed me to deepen my knowledge of the **computer vision branch of machine learning**, building directly on the experience I gained during the team project, where I was first introduced to convolutional neural networks (CNNs). The objective was to classify steel surfaces as either **acceptable or defective**, while comparing the performance of a traditional machine learning algorithm with a deep learning approach.

For the traditional machine learning model, I selected a **Support Vector Machine (SVM)** because it offers a strong balance between simplicity, interpretability, and predictive performance. For the deep learning approach, I implemented a **transfer learning model based on the ResNet50 CNN architecture**. Transfer learning enabled me to leverage knowledge learned from large pre-trained image datasets, significantly reducing both training time and computational requirements while maintaining strong predictive capabilities.

From the outset, I considered **model explainability** as a key component of the project. To ensure a fair comparison between the two approaches, I selected **occlusion sensitivity heatmaps** as a common explainable AI (XAI) technique that could be applied to both models. This enabled a direct comparison of how each model identified and utilized visual features when making predictions.

In addition to model development, I designed a broader evaluation framework to compare the two approaches beyond predictive accuracy alone. The framework incorporated factors such as **computational cost, cyclomatic complexity, robustness to lighting conditions, deployment considerations, and ethical implications**. This allowed me to assess each model holistically and determine which solution would be most appropriate for real-world implementation rather than focusing solely on performance metrics.

So What?

This assignment felt like a natural continuation of the team project, where I first gained practical exposure to CNNs. I was able to build on the lessons learned from that experience, particularly regarding **model selection, parameter tuning, and performance optimization**.

One challenge I encountered during the previous project was identifying the optimal model parameters to improve prediction performance. Drawing on that experience, I adopted a more structured approach in this project by using **k-fold cross-validation** and applying a stronger understanding of **classification thresholds**. These techniques enabled me to evaluate model performance more reliably and improve predictive accuracy while reducing the risk of overfitting.

Perhaps the most challenging aspect of the project was understanding and explaining how the models arrived at their decisions. Although occlusion sensitivity heatmaps provided valuable visual insights, interpreting them proved more difficult than expected. Initially, I assumed that regions with higher activation values would always correspond to features that strongly influenced the final prediction. However, I discovered that this relationship is more nuanced.

Through further investigation, I learned that activation values should be interpreted as indicators of the model's **attention** rather than direct measures of feature importance. A region may attract considerable attention from the model because it appears unusual or potentially relevant, yet ultimately contribute little to the final classification decision. Conversely, regions with relatively modest activation levels may still play a critical role in determining the predicted class. For example, a model may carefully examine a suspicious-looking surface region before correctly identifying it as a harmless manufacturing variation rather than a defect.

This realization significantly improved my understanding of explainable AI and reinforced the importance of interpreting visual explanations carefully. It also highlighted that explainability techniques provide valuable evidence about model behaviour, but they must be analysed within the broader context of the model's decision-making process rather than in isolation.

Now What?

Although I am highly satisfied with the overall outcome of this project, I have identified several opportunities for improvement that I intend to explore in future machine learning projects, both academically and professionally.

One limitation observed during the evaluation process was a slight dependence on **non-saturated lighting conditions**. This suggests that the models may be vulnerable to camera-specific characteristics and changes in illumination, potentially introducing unwanted bias into predictions. Improving robustness to varying lighting conditions is therefore a key area for future development.

To address this issue, I plan to expand the current image augmentation strategy by incorporating **clipped-offset and saturation-based augmentations**, which would expose the model to more examples of overexposed and highly illuminated images during training. This should improve its ability to generalize across different camera settings and environmental conditions.

Another promising avenue is the implementation of **paired-view consistency loss**. In addition to the standard classification objective, this approach introduces a penalty whenever predictions for the same object differ significantly under different lighting conditions. By encouraging prediction consistency across varying visual environments, the model can learn features that are more invariant to illumination changes and therefore more robust in real-world applications.

More broadly, this project reinforced the importance of evaluating machine learning systems using criteria beyond predictive performance. Factors such as explainability, computational efficiency,

deployment feasibility, and robustness are equally important when determining whether a model is suitable for practical use. Moving forward, I intend to adopt this more holistic perspective when designing and evaluating machine learning solutions.

Overall, this project strengthened my expertise in **computer vision, transfer learning, model evaluation, and explainable AI**. It also deepened my understanding of the trade-offs that exist between traditional machine learning and deep learning approaches. Most importantly, it demonstrated that successful machine learning projects require not only strong predictive performance but also careful consideration of interpretability, robustness, and real-world applicability.